

Advisor update: LatentMAS relay compression

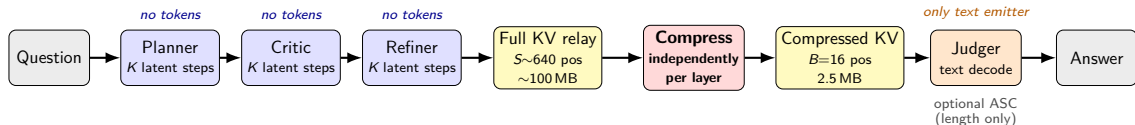
Week of 2026-07-30 – 2026-08-05 · Qwen3-14B

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August 2026

Verified LatentMAS fork · Qwen3-14B · greedy decode unless noted

Architecture



Compress($C; B=16, s=4, r=8$) — every layer

- 1 Score position i : $\sigma_i = \sum_{h=1}^{n_{kv}} \|K_{h,i}\|_2$ (key-norm; alt: value-norm / recency)
- 2 Keep sinks $\{0, \dots, s-1\}$ **always**, plus Top_{B-s} by σ among $i \geq s$
- 3 Sort kept indices ascending; gather those K/V slices $\rightarrow$ length B
- 4 **evict**: done. **obf**: split dropped into r ordered buckets, mean-pool, append $\Rightarrow$ length $B+r$
- 5 Rebuild HF DynamicCache; Judger attends only to this prefix

One layer: keep vs drop



■ sink ■ top- σ ■ drop



output length: 16 (evict) or 24 (obf)

GSM8K: $640 \rightarrow 16 = 40\times$ KV

Setup in one slide

Question → Planner → Critic → Refiner → **relay KV** → Judger (text) → Answer

- Upstream agents reason in latent space (K steps, no tokens) and grow a shared KV cache.
- Only the Judger decodes. Relay is typically $\sim 640\text{--}1100$ positions / $\sim 100\text{--}180$ MB.
- Prior finding: relay is largely **content-insensitive within task** (wrong-question $\approx$ right-question) $\Rightarrow$ should compress.

This week's question

Can we shrink the relay before the Judger, keep accuracy, and what does that cost?

Method (what we built)

Compressor (relay_compress.py)

- Keep first $s=4$ sinks + top- $(B-s)$ by key-norm
- Optional OBF: mean-pool evicted mass into $r=8$ slots
- Modes: full / evict / obf

Harness (exp_2x2_relay_asc.py)

- Build upstream once $\rightarrow$ compress $\rightarrow$ batched Judger
- Arms A-F + shuffled / none / crosstask
- Optional Judger-only ASC (SEAL @ L28)

2x2

	no ASC	ASC
full	A	B
obf	C	D

- + E/F = plain evict
- + S/N/X diagnostics

Headline: GSM8K $\approx 40\times$ at iso-acc

Arm	Acc	Relay MB	Positions
A full	0.817	100.0	~ 640
E evict @16	0.817	2.5	16
C obf @16 (+r)	0.800	3.75	24

- Replicated on seeds **42 / 123 / 7** ($n=60$ each).
- Plain eviction matches full; mean-pool OBF backfill not needed here.
- Shuffled $\approx$ full (0.800 vs 0.817) — scaffold story still holds.

Claim-ready

GSM8K: cut inter-agent KV residency $\sim 40\times$ with no accuracy loss.

Harder tasks: thicker residual required

MATH (full 0.783)

Keep B	Acc	vs full
128	0.800	+2
64	0.733	-5
32	0.717	-7

MedQA (full 0.633)

Keep B	Acc	vs full
128	0.567	-7 soft
64	0.517	-12
16-32	~0.36	collapse

Operating rule

Easy tasks tolerate extreme shrink; harder tasks need keep ~ 64 –128.

MedQA with **no** cache: 0.267 — presence still matters, thickness matters more.

Cost: compression overhead is real, but tiny

$$t_{e2e} = t_{\text{upstream}} + t_{\text{compress}} + t_{\text{decode}}$$

- GSM8K keep-16: $\Delta t_{\text{decode}}(C-A) \approx -1.05 \text{ s}$, $\Delta t_{e2e} \approx -1.02 \text{ s}$
- $\Rightarrow t_{\text{compress}} \approx \mathbf{20\text{--}25 \text{ ms}}$ ($\sim 0.7\%$ of upstream $\sim 3 \text{ s}$)
- Peak GPU: $36.8 \rightarrow 34.1 \text{ GB}$; relay residency: $100 \rightarrow 2.5 \text{ MB}$

Honest accounting

$40\times$ is **relay / Judger-prefix memory**, not “skip upstream FLOPs.”

We still build the full cache once; then shrink it.

AIME 2024 — long runs, what we learned

- Full LatentMAS on all 30 AIME problems, up to 8192 output tokens (slow: many minutes per K , re-run overnight to confirm).

Latent steps K	Accuracy	Mean tokens	Mean latency
0 (Judge only)	0.53	7237	178 s
10	0.67	5927	146 s
40	0.50	6401	159 s

- Latent agents help on hard math: +14 pts at $K=10$ vs Judge-only (overnight re-run: 0.67 vs 0.57).
- More latent steps is not better — $K=40$ falls back to $\sim$ Judge-only.
- Cutting K does **not** save wall-clock; Judge length dominates.

Compression on AIME

Tried, but answers often hit the 8192 cap so scores are not trustworthy yet. Compression claims stay on GSM8K / MATH / MedQA for now.

Status and proposed next

This week

- GSM8K: $\sim 40\times$ relay compression at matched accuracy (3 seeds)
- MATH / MedQA: mapped how aggressive we can be before accuracy drops
- AIME: confirmed LatentMAS helps at $K=10$; compression deferred

Next (pick)

- 1 Put these numbers into the paper draft
- 2 Memory-accuracy Pareto figure across keep-budgets
- 3 AIME compression only if we re-run without truncation issues